PROJECT CODEBOOK

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Industry: Healthcare
Project Title: Improving Maternal Health Insights Through Data Analysis
Proposed Approach: Python-Based Analysis of Maternal Health Data
Dataset: NGIR8BFL.DTA (Nigeria Demographic and Health Survey 2024 Individual Recode).
Analysis Domain: Respondents with valid recorded responses for the outcome being analyzed.

Source Variable	Analysis Variable	Description	Coding / Treatment
caseid	caseid	Respondent identifier	Unique respondent/case identifier; retained for record identification, not used as a numeric measure.
v012	age	Respondent age	Age in completed years; numeric.
v025	residence	Place of residence	1 = Urban; 2 = Rural.
v106	education	Highest educational level	0 = No education; 1 = Primary; 2 = Secondary; 3 = Higher.
v190	wealth_group	Household wealth index	1 = Poorest; 2 = Poorer; 3 = Middle; 4 = Richer; 5 = Richest.
m1_1	tetanus_category	Number of tetanus injections before birth	Code 8 (“Don’t know”) set to missing. Code 0 = No injections; 1–6 retained as counts; 7 = 7 or more. In the analysis extract, 214 records had this item missing while other core maternity items were present.
m14_1	antenatal_visits	Number of antenatal visits during pregnancy	Code 98 (“Don’t know”) set to missing; Valid numeric counts retained. Used to estimate the survey-weighted mean.
m15_1	delivery_group	Place of delivery	11–12 = Home; 21–27 = Public facility; 31–38 = Private facility; 41, 42, 46 = NGO facility; 96 = Other. Group-header codes are not respondent categories.
m17_1	c_section	Delivery by caesarean section	0 = No; 1 = Yes. Other/non-response values are not recoded as Yes or No.
v005	weight	Individual sample weight	Divided by 1,000,000 to create the analysis weight.
v021	cluster	Primary sampling unit (PSU)	Cluster identifier used in survey variance estimation.
v022	stratum	Sample stratum	Stratum identifier used in survey variance estimation.

Derived Analysis Fields and Interpretation Notes

- **age:** Numeric age in years, derived from v012.
- **residence, education, wealth_group:** Readable categorical labels derived from v025, v106, and v190 respectively.
- **tetanus_category:** Categorical presentation of m1_1; 7 is top-coded as "7 or more."
- **antenatal_visits:** Numeric count from m14_1 after converting code 98 to missing.
- **delivery_group:** Grouped delivery location from m15_1 using the categories in the table.
- **c_section:** Readable Yes/No response derived from m17_1.
- **weight, cluster, stratum:** Survey-design fields derived from v005, v021, and v022. Used with Taylor linearization in Samplics 0.6.0.

Missing Data and Denominator Note.

- Codes 8 (m1_1) and 98 (m14_1) represent "Don't know" and were set to missing.
- Missing maternity responses were not automatically treated as data errors because the maternity questions do not apply to every respondent.
- Estimates use the valid-response domain for each outcome; therefore, denominators may differ.

Survey Design.

- Weight = v005 / 1,000,000;
- PSU = v021;
- Stratum = v022.
- Weighted means and proportions are reported
- Weights are not used here to produce population totals.